

Course Title	Digital Logic Design		
Course Code	MC116	Credit Structure	1-0-2-2 (L-T-P-C)
Category	Core	Semester	I - Autumn Semester (AY 25-26)
Program	B.Tech(MnC)		
Prerequisites	None		
Course Objectives/ Brief Course Description	This course provides an introduction and basic understanding of the design and implementation of digital circuits. Topics include number systems, Boolean algebra, combinational and sequential circuits.		
Evaluation/ Grading Policy	• Mid-Semester Exam: 30% • End-Semester Exam: 30% • Laboratory Assignments: 10% • Quizzes / Class Participation: 10% • Project / Mini Design Task: 10%		
Course Materials/ References	Basic understanding of digital and analog domain, noise margin, number system, binary code, logic gate and related devices, Boolean algebra and simplification Combinational Circuits: Digital arithmetic, encoders, decoders, multiplexer, de-multiplexers Sequential Circuits: Latch, flip-flop, register, counter, memory.		
Detailed Course Content	APPENDIX A: Lectures/Modules APPENDIX B: Lab Experiments		

Course Outcomes:

- **CO1:** Understand the representation of data using number systems and binary codes.
- **CO2:** Apply Boolean algebra and simplification techniques to design logic circuits.
- **CO3:** Design and implement combinational logic circuits using basic gates and ICs.
- **CO4:** Analyze and build sequential logic systems like flip-flops, counters, and registers.
- **CO5:** Use digital ICs and breadboard setups to verify logical functions through lab experiments.

POs-COs Matrix:

	P1	P2	P3	P4	P5	P6–P12
CO1	✓					
CO2	✓	✓				
CO3	✓	✓	✓			
CO4	✓	✓				
CO5	✓	✓				

CO#	Course Outcome	Mapped POs	Justification
CO1	Understand the representation of data using number systems and binary codes	P1	Fundamental knowledge of engineering concepts and mathematics
CO2	Apply Boolean algebra and simplification techniques to design logic circuits	P1, P2	Involves problem analysis and use of engineering fundamentals
CO3	Design and implement combinational logic circuits using basic gates and ICs	P1, P2, P3	Involves design and development of solutions
CO4	Analyze and build sequential logic systems like flip-flops, counters, and registers	P1, P3	Analyzing and designing real-world sequential systems
CO5	Use digital ICs and breadboard setups to verify logical functions through labs	P4, P5	Emphasizes experimentation, practical tools, and techniques

APPENDIX A: Detailed Course Content (Session-wise/ Module-wise) – Tentative Plan

Week	Module / Topic	Description	Lectures
1	Introduction to Digital Systems	Analog vs Digital, signals and domains, noise margin	1
2	Number Systems	Binary, octal, hexadecimal, conversions	1
3	Binary Codes and Logic Gates	Binary codes, basic and universal logic gates	1
4	Boolean Algebra	Postulates, theorems, simplification	1
5	Logic Simplification	Karnaugh Maps, SOP/POS forms	1
6	Combinational Circuits I	Half adder, full adder, subtractors	1
7	Combinational Circuits II	Encoders, decoders	1
8	Combinational Circuits III	Multiplexers, demultiplexers	1
9	Sequential Circuits I	Latches and Flip-Flops	1
10	Sequential Circuits II	Registers and Counters	1
11	Memory Devices	RAM, ROM, basics of memory hierarchy	1
12	Lab Demonstrations / Projects	Practical implementation of combinational and sequential logic	1

APPENDIX B: Lab Assignments (Session-wise) – Tentative Plan

Lab 1: Familiarization of Various Logic Gates

Test AND, OR, EXOR gates on breadboard. Implement one's complement of a 4-bit number A when control input C=1, else leave A unchanged.

Lab 2: Binary Arithmetic and Binary Codes

Implement half adder and half subtractor. Design a 4-bit binary to Gray code converter and even parity generator.

Lab 3: Four-bit Increment and Decrement Circuits

Implement 4-bit decrementer with control input. Derive expression and design for 4-bit increment-by-2 circuit.

Lab 4: Universal Logic Gates

Implement given function using only NAND gates. Design $F(A,B,C) = \sum m(2,3,4,5)$ using 2 NAND ICs.

Lab 5: Magnitude Comparators

Design and implement 3-bit equality checker and 2-bit greater-than comparator.

Lab 6: Study of Adder/Subtractor and Up/Down Counter

Use CD4008 and XOR IC for 4-bit adder/subtractor. Study counter transitions using 4029 IC.

Lab 7: Familiarization with Multiplexer and Decoder ICs

Use 8:1 MUX to implement full adder sum function and given logic function. Verify 3-to-8 decoder (79LS138).

Lab 8: Design of Counters and Flip-Flop Conversion

Verify D and JK flip-flops. Construct SR from D flip-flop. Design a 3-bit JK counter from 000 to 111.

Programme Outcomes (POs)

PO No.	Programme Outcomes
PO1	Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
PO2	Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences
PO3	Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
PO4	Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
PO5	Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
PO6	The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
PO7	Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO8	Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
PO9	Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
PO10	Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
PO11	Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
PO12	Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.